

Data Structures Lab (CL2001)

Date: October 16th 2025

Instructor(s)

Mr. Muhammad Kamran

Sessional Exam

Total Time (Hrs): 1.5

Total Marks: 15

Total Questions: 2

Roll No

Section

Student Signature

Do not write below this line

Attempt all the questions.

CLO #: 1

Q1: [7 Marks]

Write a recursive function to determine whether a singly linkedlist, given its head, is palindrome.

Input: A -> B -> A -> B -> A -> nullptr

Output: true

Input: A -> B -> A -> b -> a -> nullptr

Output: true

Input: M -> O -> S -> C -> O -> W -> nullptr

Output: false

CLO #: 2

Q2: [8 Marks]

- a) [7 Marks] You are given the head pointer of a singly linked list. Your task is to rearrange the linked list in such a way that all even-positioned nodes (2nd, 4th, 6th, ...) are enqueued first, followed by all odd-positioned nodes (1st, 3rd, 5th, ...), while preserving their relative order within each group. However, you are not allowed to create a new linked list. Instead, you must use a queue as a logical helper to simulate the reordering process.

Original Linked List:

10 → 20 → 30 → 40 → 50 → 60 → 70 → NULL

Rearranged Linked List:

20 → 40 → 60 → 10 → 30 → 50 → 70 → NULL

- b) [1 Mark] Create a Menu for all questions